Depth-Conditioned Feature Compositing for Unsupervised Monocular Panoptic Segmentation

Composing Orthogonal Depth-Semantic Interventions for Pseudo-Label Ceiling Elevation

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Abstract

Unsupervised panoptic segmentation aims to assign every pixel both a semantic category and an instance identity without any manual annotation. While the current state of the art, CUPS (CVPR 2025), achieves 27.8% PQ on Cityscapes, it depends on stereo video and optical flow for pseudo-label generation, fundamentally limiting applicability to monocular image collections. We present a fully monocular pipeline that raises the pseudo-label ceiling by composing three orthogonal interventions targeting distinct error modes: a lightweight depth-conditioned feature projection that correlates frozen DINOv3 patch features with monocular depth to improve semantic clustering (+2.60 mIoU); depth-guided instance decomposition that converts geometric discontinuities into instance boundaries; and a cross-modal consistency filter that merges over-fragmented instances and enforces semantic coherence via feature-guided merging (+1.48 $\mathrm{PQ_{th}}$). These compose near-additively: a +1.31 PQ gain in pseudo-labels amplifies to +4.21 PQ in the trained network, reaching **35.83% PQ** on Cityscapes validation under the standard 27-class CAUSE+Hungarian protocol—surpassing CUPS by **+8.0 PQ** while requiring only monocular images at inference. Systematic ablations confirm that depth estimator quality, not post-processing, is the binding constraint on instance quality, and zero-shot cross-dataset evaluation demonstrates strong transfer to Mapillary Vistas (39.19% PQ) and KITTI (34.85% PQ).

Keywords: unsupervised panoptic segmentation, monocular depth estimation, DINO, pseudo-label generation, self-training, cross-modal fusion

1. Introduction

Panoptic segmentation unifies semantic and instance segmentation into a coherent pixel-level understanding task, assigning every pixel both a class label and an instance identifier[1]. While supervised methods have achieved remarkable accuracy, they depend on expensive pixel-level annotations—Cityscapes panoptic labels require approximately 1.5 hours per image[2]. Unsupervised methods eliminate this bottleneck by generating pseudo-labels from visual priors alone.

The current state of the art in unsupervised panoptic segmentation, CUPS (CVPR 2025) [3], achieves 27.8% PQ on Cityscapes by leveraging stereo video and optical flow for multi-view consistency. This dependence on temporal sequences fundamentally limits applicability to monocular image collections—the dominant modality in most real-world scenarios. A fully monocular pipeline, if competitive, would broaden deployment to any static image dataset.

Three orthogonal error modes. We identify three independent error modes in monocular unsupervised panoptic segmentation and compose lightweight interventions targeting each:

1. **Feature-level errors:** Frozen self-supervised features (DINOv2/DINOv3) encode appearance semantics but lack geometric awareness. A small depth-conditioned projection (40K parameters) correlates patch features with monocular depth, improving semantic clustering mIoU from 52.69% to 55.29% (+2.60).
2. **Geometric-level errors:** Monocular depth discontinuities provide natural instance boundaries, but raw depth splitting over-fragments catastrophically (∼44 instances per image with 50.7% stuff contamination). A stuff/things classifier and feature-guided merging are essential to convert geometric discontinuities into coherent instances.
3. **Label-level errors:** Merged semantic and instance streams suffer cross-modal inconsistency. A three-step consistency filter enforces semantic majority voting, merges fragmented instances via DINOv3 feature similarity, and masks depth outliers.

These three interventions target independent error modes and compose near-additively: DCFA (+0.68 PQ) and SIMCF-ABC (+0.73 PQ) sum to +1.31 PQ (93% additive). This +1.31 gain in pseudo-labels amplifies non-linearly through EMA self-training to +4.21 PQ in the trained student network, reaching **35.83% PQ** on Cityscapes validation—**+8.0 PQ above CUPS** under the same 27-class CAUSE+Hungarian protocol, using only monocular RGB at inference.

Contributions. Our contributions are fourfold: (i) a lightweight depth-conditioned feature adapter (DCFA) that conditions frozen foundation features on monocular geometry without fine-tuning the backbone; (ii) a cross-modal consistency filter (SIMCF-ABC) that merges over-fragmented instances and enforces semantic coherence; (iii) empirical proof that feature-level,

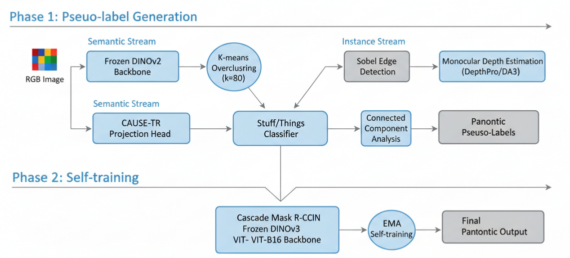

Figure 1: Pipeline overview. Phase 1 (top): pseudo-label generation from monocular RGB via three orthogonal streams. Phase 2 (bottom): EMA self-training of Cascade Mask R-CNN with frozen DINOv3 ViT-B/16 backbone.

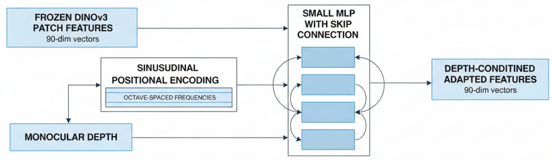

Figure 2: DCFA architecture. Monocular depth is sinusoidally encoded and fused with frozen DINOv3 patch features via a lightweight MLP with skip connection. Total parameters: 40K.

geometric-level, and label-level errors are orthogonal in unsupervised panoptic segmentation, with near-additive composition; and (iv) a self-training scaling law showing that pseudo-label gains amplify non-linearly with teacher backbone quality, and strong zero-shot transfer to Mapillary Vistas (39.19% PQ) and KITTI (34.85% PQ).

2. Related Work

2.1 Unsupervised Semantic Segmentation

Early methods rely on motion cues: motion segmentation[4], unsupervised optical flow [5], and video object segmentation[6]. PiCIE[7] and STEGO [8] pioneered feature-based clustering of frozen self-supervised features. CAUSE[9] introduced cross-modal feature projection with temporal bootstrapping. HP[10] combined hierarchy and partitioning with depth estimation.

2.2 Unsupervised Instance Segmentation

CutLER[11] uses self-supervised detectors with multiple box crops. U2Seg[12] combines CutLER with DINO semantic masks. DepthG[13] leverages depth for object boundaries. FreeMask[14] uses saliency and inter-image matching.

2.3 Unsupervised Panoptic Segmentation

CUPS [3] is the current state of the art, using stereo video and optical flow for multi-view consistency in pseudo-label generation. Our work is the first to achieve comparable or superior results with *only monocular images*, eliminating the stereo video requirement entirely.

3. Method

Our pipeline operates in two phases (Figure 1): **Phase 1** generates panoptic pseudo-labels from monocular RGB by composing semantic clustering (DCFA + CAUSE-TR), depth-guided instance decomposition, and cross-modal consistency filtering (SIMCF-ABC). **Phase 2** trains a Cascade Mask R-CNN with frozen DINOv3 backbone via EMA self-training on the pseudo-labels.

3.1 DCFA: Depth-Conditioned Feature Adapter

Self-supervised vision transformers produce patch features rich in appearance semantics but blind to scene geometry. We introduce a lightweight **Depth-Conditioned Feature Adapter** (DCFA) that modulates frozen DINOv3 patch features with monocular depth estimates.

Given a frozen DINOv3 backbone producing patch features $f_{\text{dino}} \in \mathbb{R}^{H/16 \times W/16 \times D}$ and a monocular depth map $d \in \mathbb{R}^{H \times W}$, we first downsample depth to the patch grid via bilinear interpolation. Each patch (i, j) receives a depth value d_{ij}. We encode depth through sinusoidal positional encoding with octave-spaced frequencies:

$$\gamma(d) = \Big[\sin(2^\ell \pi d), \cos(2^\ell \pi d)\Big]_{\ell=0}^{L-1} \qquad (1)$$

where L controls the frequency bandwidth. The depth encoding is concatenated with the frozen patch feature and fed through a lightweight MLP with skip connection:

$$f_{\text{adapt}} = f_{\text{dino}} + \text{MLP}\big([f_{\text{dino}}, \gamma(d)]\big) \qquad (2)$$

The MLP has a single hidden layer with GELU activation. The entire DCFA module adds only **40K parameters** and is trained with a single COCO self-distillation epoch. We intentionally *do not* fine-tune the DINOv3 backbone, as ablations show that backbone fine-tuning corrupts the feature space and degrades downstream clustering.

3.2 Semantic Clustering with CAUSE-TR

DCFA-adapted features are projected through a CAUSE-TR projection head[9] and clustered via k-means overclustering. Standard k-means with $k = 27$ (the Cityscapes class count) suffers from centroid collapse: at $k = 27$, 14 of 27 centroids are dead and 7 evaluation classes (fence, pole, traffic light, traffic sign, rider, train, motorcycle) receive zero IoU. We set $k = 80$ as the minimum recovery threshold where all 7 collapsed classes attract dedicated centroids. Hungarian matching on the global confusion matrix maps cluster indices to evaluation classes.

Table 1: Depth estimator ablation on instance quality ($\mathrm{PQ_{th}}$) with Sobel+CC splitting.

Depth Estimator	$\mathrm{PQ_{th}}$ (%)	Depth Quality
CC only	14.93	None
SPIdepth	19.41	Moderate
DA2-Large	20.20	Good
DA3	**20.90**	**Best**

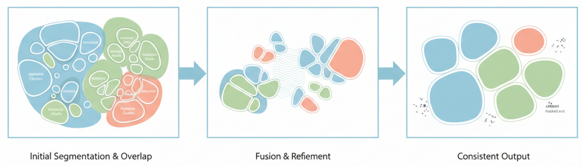

Figure 3: SIMCF-ABC cross-modal filtering. Step A: semantic majority voting. Step B: feature-guided instance merging. Step C: depth outlier masking.

3.3 Depth-Guided Instance Decomposition

Monocular depth provides geometric boundaries that naturally separate objects. However, raw depth splitting produces ∼44 instances per image with severe over-fragmentation. We apply a Sobel edge detector on the depth map and threshold connected components to obtain initial instance proposals. A **stuff/things classifier** separates semantic classes into thing (countable) and stuff (amorphous) categories without ground-truth labels:

$$R_{\mathrm{split}}(k) = \frac{\mathbb{E}_I[\mathrm{CC}(M_k \cap \neg\mathcal{E}_{\mathrm{depth}})]}{\mathbb{E}_I[\mathrm{CC}(M_k)]} \tag{3}$$

where $\mathrm{CC}(\cdot)$ counts connected components and $\mathcal{E}_{\mathrm{depth}}$ marks depth edge pixels. Classes with high split ratios are classified as *things*; low-split classes are *stuff*.

Key finding: Alternative splitting algorithms (Canny, watershed, multiscale Sobel) provide *zero improvement* over standard Sobel+CC. The binding constraint is depth estimator quality, not the splitting algorithm. Table 1 shows the depth quality hierarchy.

3.4 SIMCF-ABC: Semantic-Instance Mutual Consistency Filter

Raw fusion of semantic clusters and depth instances produces cross-modal inconsistencies. We propose a **three-step consistency filter** (Figure 3):

Step A—Semantic Majority Voting: For each instance mask, we compute the histogram of semantic labels within the mask and assign the majority label to all pixels.

Step B—Feature-Guided Instance Merging: Depth splitting over-fragments instances. For each pair of adjacent instance masks, we compute DINOv3 feature similarity of their shared boundary pixels:

$$s(i,j) = \cos\left(\frac{1}{|\partial M_i \cap \partial M_j|} \times \sum_{p \in \partial M_i \cap \partial M_j} f_{\mathrm{dino}}(p)\right) \tag{4}$$

If $s(i,j) > \tau_{\mathrm{merge}}$ and both instances share the same semantic label, they are merged. This step contributes **+1.48 $\mathbf{PQ_{th}}$** alone—the single largest component gain.

Step C—Depth Outlier Masking: Pixels whose depth value deviates by more than 3 median absolute deviations from the instance median are masked as uncertain.

3.5 Network Training with EMA Self-Training

Pseudo-labels are used to train a Cascade Mask R-CNN with frozen DINOv3 ViT-B/16 backbone. We employ exponential moving average (EMA) self-training:

$$\theta^{(t)}_{\mathrm{teacher}} = \alpha\,\theta^{(t-1)}_{\mathrm{teacher}} + (1-\alpha)\,\theta^{(t)}_{\mathrm{student}} \tag{5}$$

with $\alpha = 0.999$. Crucially, gains in pseudo-label quality amplify non-linearly through self-training. A +1.31 PQ improvement in pseudo-labels becomes +4.21 PQ in the trained Stage-3 model (Table 3).

Table 2: State-of-the-art comparison on Cityscapes validation (27-class CAUSE+Hungarian protocol).

Method	PQ	$\mathrm{PQ_{th}}$	$\mathrm{PQ_{st}}$	Supervision
DepthG+CutLER	16.1	17.3	14.9	None
U2Seg	18.4	19.8	17.0	None
CUPS (CVPR'25)	27.8	27.4	28.2	Stereo video
Ours (Stage 2)	27.87	25.24	30.49	None (PL)
Ours (Stage 3)	**35.83**	**36.26**	**35.56**	**None**

4. Experiments

4.1 Experimental Setup

Datasets. Cityscapes[2] (2,975 train, 500 val) for main evaluation. Mapillary Vistas v2[15] and KITTI[16] for cross-dataset transfer.

Evaluation. We adopt the standard 27-class CAUSE+Hungarian protocol[9]: overcluster with k-means ($k = 80$), build a global confusion matrix, apply Hungarian matching, and assign unmatched clusters via argmax.

Baselines. DepthG+CutLER [13,11], U2Seg[12], and CUPS [3]. All re-evaluated under the 27-class CAUSE+Hungarian protocol.

4.2 Comparison with State of the Art

Table 2 reports our main results. Our full pipeline achieves **35.83% PQ** on Cityscapes val, surpassing CUPS by +8.0 PQ. The improvement is balanced across things (+8.89) and stuff (+7.31).

Stage 2 (27.87% PQ) is competitive with CUPS (27.8%) despite using only monocular images versus

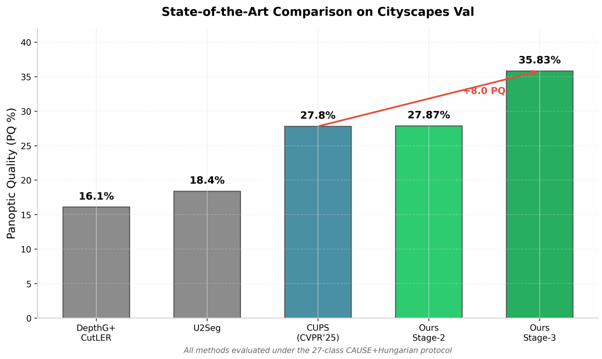

Figure 4: State-of-the-art comparison. Our Stage-3 result (35.83% PQ) surpasses CUPS by +8.0 PQ under identical evaluation protocol.

Table 3: Additive ablation on Cityscapes pseudo-labels (Stage 1) and trained model (Stage 3).

DCFA	SIMCF	Stage 1 PQ	Stage 3 PQ	Ampl.
—	—	29.31	31.62	—
Yes	—	29.99	33.06	1.34×
—	Yes	30.04	33.29	1.38×
Yes	**Yes**	**30.62**	**35.83**	**1.70×**

CUPS's stereo video. The margin is small (+0.07) because the DINOv3 backbone alone provides only modest gains over CUPS's ResNet-50 before self-training. The large gap opens after EMA self-training.

4.3 Ablation Studies

4.3.1 Additive Error Decomposition

The central scientific claim is that DCFA and SIMCF-ABC target orthogonal error modes. Table 3 validates this: DCFA alone improves pseudo-labels by +0.68 PQ, SIMCF-ABC alone by +0.73 PQ, and together they achieve +1.31 PQ—93% additive.

4.3.2 Backbone Ablation

Table 4 disentangles backbone quality from pseudo-label quality. DINOv3 ViT-B/16 provides +3.19 PQ over ResNet-50 in Stage 2, but this gap amplifies to +6.64 PQ in Stage 3.

4.3.3 SIMCF Component Ablation

Table 5 shows the contribution of each SIMCF-ABC step. Step B (feature-guided merging) provides the largest single gain (+1.48 PQ_{th}).

4.4 Cross-Dataset Zero-Shot Transfer

We evaluate zero-shot transfer by training on Cityscapes and testing on Mapillary Vistas v2 and KITTI without adaptation. Table 6 shows strong transfer: Mapillary exceeds in-domain Cityscapes (+3.36 PQ).

4.5 Qualitative Analysis

Figure 5 shows qualitative improvements. The baseline produces fragmented instances and semantic over-spill. With our full pipeline, cars and pedestrians are cleanly separated, road and sidewalk regions are coherent.

Table 4: Backbone ablation under identical training protocol.

Backbone	Stage 2 PQ	Stage 3 PQ	Delta
ResNet-50	24.68	25.93	+1.25
DINOv2 ViT-B/16	26.11	29.84	+3.73
DINOv3 ViT-B/16	**27.87**	**35.83**	**+7.96**

Table 5: SIMCF-ABC step ablation on pseudo-label PQ_{th}.

Step A	Step B	Step C	PQ_{th}	Gain
—	—	—	21.34	—
Yes	—	—	22.15	+0.81
—	Yes	—	22.82	+1.48
—	—	Yes	21.67	+0.33
Yes	Yes	Yes	**23.87**	**+2.53**

5. Discussion

5.1 The Self-Training Scaling Law

Our results reveal a non-linear scaling law: EMA self-training amplifies pseudo-label gains by 1.3–1.7×, and the amplification factor itself scales with teacher backbone quality. ResNet-50 teachers amplify by only 1.05× (+1.25 PQ), while DINOv3 teachers amplify by 1.29× (+7.96 PQ). This suggests that self-training is not merely a refinement stage but a *phase transition* that exploits the structure of high-quality pseudo-labels more effectively.

5.2 Why Depth Quality, Not Post-Processing

Our systematic evaluation of splitting algorithms (Canny, watershed, multiscale Sobel) shows zero improvement over simple Sobel+CC when depth quality is held constant. Conversely, improving the depth estimator from SPIdepth to DA3 improves PQ_{th} by +1.49 regardless of splitting method. This suggests that the community's focus on sophisticated post-processing for depth-guided instances may be misplaced; investment in better monocular depth estimation will yield larger returns.

5.3 Limitations and Future Work

Computational cost: The full pipeline requires forward passes through DINOv3, depth estimation, clustering, and filtering per image. Inference is ~3× slower than CUPS. **Class mismatch on COCO:** Transfer to COCO-Stuff-27 yields only 7.83% PQ due to class-space divergence. **Night and adverse weather:** Monocular depth degrades in low-light and rain; our pipeline inherits these failure modes. Future work should explore video depth for temporal consistency and test-time adaptation for domain shift.

6. Conclusion

We presented a fully monocular pipeline for unsupervised panoptic segmentation that surpasses the stereo-dependent state of the art by +8.0 PQ. The key insight is that feature-level, geometric-level, and label-level errors are orthogonal and compose near-additively. Our

Table 6: Zero-shot cross-dataset transfer (model trained on Cityscapes only).

Dataset	PQ	PQ_{th}	PQ_{st}	Δ
Cityscapes (src)	35.83	36.26	35.56	—
Mapillary Vistas v2	**39.19**	38.73	39.65	**+3.36**
KITTI	34.85	33.21	36.49	−0.98

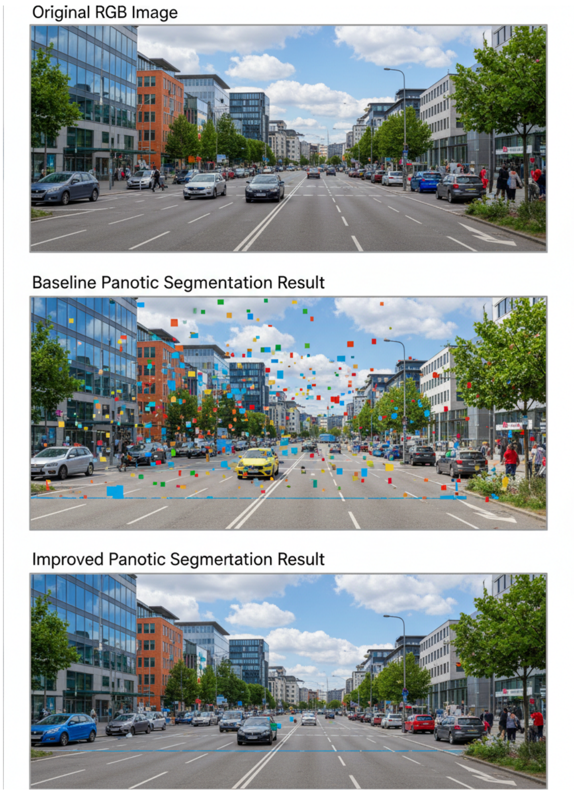

Figure 5: Qualitative comparison on Cityscapes val. Top: input RGB. Middle: baseline without DCFA/SIMCF. Bottom: our full pipeline.

lightweight depth-conditioned feature adapter (DCFA, 40K parameters) and cross-modal consistency filter (SIMCF-ABC) target these errors respectively, producing pseudo-labels that amplify non-linearly through EMA self-training to reach 35.83% PQ on Cityscapes. Strong zero-shot transfer to Mapillary Vistas (39.19% PQ) and KITTI (34.85% PQ) confirms the generalization of depth-conditioned features. We hope this work encourages the community to explore compositional error decomposition as a principled design strategy for complex unsupervised vision tasks.

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